

TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS

IFA	2	18.06.24	REVISED WHERE INDICATED IN RED	ITF (FDT)	ITF (IC)	SEME
IFA	1	08.11.22	ISSUED FOR APPROVAL	TAM (MT)	ITF (IC)	SEME
IFA	0	01.07.22	ISSUED FOR APPROVAL	TAM (LD)	ITF (IC)	SEME
Status	Rev. Num.	Date	Description	Prepared by	Approved by	Received by
Rev. Index						
Client  SOUND ENERGY Exploration & Production SOUND ENERGY Mahaj Riad Center, Batiment 6, 3eme etage, Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco				TENDRARA PROJECT		
Contractor  ITALFLUID MOROCCO BRANCH Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca				Provision of a leased train for production of LNG for Sound Energy		
Engineering  ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) ITALY Tel:085-4224374 / fax: 085-4220742						
Technology Vendor   TECHNICAL AMERICA INC 301 N Smith Avenue, 92880 Corona (CA)-USA Ph. +1 951 272 9540						
Facility Name TENDRARA			Location MOROCCO	Scale A4	Sheet of sheets 1/45	
Document Title PROCESS SAFETY AND ISOLATION PHILOSOPHY				Doc. Code SAF01-ING-FR01		
				Vendor Doc. code 21044-PR-PSP-00001		

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	Contractor ITALFLUID. MOROCCO	Vendor Document Code 21044-PR-PSP-00001		
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1. INTRODUCTION

1.1. PROJECT DESCRIPTION

Italf fluid group intends to develop the Raw Gas Treatment and LNG Production Plant for Sound Energy, to be installed in Tendirara. Technical America Inc is selected by Italf fluid group as Technology Vendor.

All provided units are designed on a “skid-mounted” philosophy to the fullest extent possible; this means that every unit will be supplied with a skid frame, with local instruments terminated at local junction boxes, with process piping and isolation/safety/process control valves and connection flanges located at skid edge for interconnecting piping to headers, sub headers or other units.

The installation inlet battery limit is intended at the flange of the delivery manifold located at plant's entrance. The outlet battery limit is intended:

- At loading point of produced LNG (hoses included)
- At evaporation pond of the disposed process water (by ITF)

This document contains general and specific information and requirements on the proposed processing Facility.

Feed natural gas at regulated pressure enters a separator for removal of entrained liquid droplets. The stream is then purified in a pre-treatment system designed to prevent freeze-up of water and carbon dioxide in the low-temperature section of the plant. The pre-treatment system has two main units, one unit to remove CO₂ (UNIT 330 – Amine Sweetening) using regenerable absorption system with amine and the other to remove water (UNIT 310 - Dehydration) using a pre-cooler and a Thermal Adsorption system using molecular sieve. Captured CO₂ (a stream with 89% mol plus CO₂) will be vented to atmosphere. UNIT 330 requires as utilities hot oil, UNIT 310 requires refrigerant R1 and hot oil.

From UNIT 200 (LNG Plant inlet) and specifically from the three phase separator V-201 we separate the natural gas from condensate and process water (process oily water). The condensate contains heavy hydrocarbons and is used in the hot oil package.

From UNIT 310 & 330 a condensate (mostly process water) is also produced, the process water is sent to the Process water treatment (UNIT 530).

The purified feed gas is cooled using a mixed refrigerant closed type refrigeration loop.

To avoid possibility of heavy hydrocarbons freezing in the cold box, a condensate stream is separated at a temperature of -75°C in a first cooling stage.

As per design basis heavy hydrocarbon are not present in the gas stream, so this separation is, in theory, not required for this project. Anyhow, considering field gas composition can change overtime and based on our experience a heavy condensate separation section was included to avoid possible operational trouble. This condensate can be warmed and vaporized with hot oil and used as fuel gas.

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The gas is then further cooled and liquefied in the cold box. The liquified natural gas exiting from the cold box is at high pressure and is sent to a Nitrogen Rejection Unit to strip out the excessive Nitrogen dissolved in the LNG (project requirements is to have a maximum N₂ content in the LNG of no more than 1.1%mol). The LNG pressure drops to the NRU operating pressure and before going to the low pressure storage is further expanded to a lower pressure in V-221 (LNG separator). The LNG flow rate from V-221 to the storage tank is measured for overall plant performance calculation, also the quality of the LNG is monitored using a gas analyser.

The liquefied natural gas is then sent to the storage tank at low pressure. All the flash gas produced during the process, the boil off gas from the LNG storage tank and LNG loading unit is recovered by a BOG compressor that boosts the pressure of this gas to fuel gas network.

The fuel gas is used in the power generation unit and hot oil unit.

Part of the flash gas that exceeds the fuel network requirements can be returned to the process upstream UNIT 310 using a high pressure second stage flash/boil off gas recycle compressor (S-362). This gas can be used intermittently as regeneration gas to regenerate the spent molecular sieve in UNIT 310.

A cryogenic nitrogen rejection unit is included to separate N₂ from the produced LNG. The vent stream from the NRU can be sent to the fuel system.

The fuel gas network and BOG/flash gas compressor are capable of auto balancing the produced flash gas depending on the operating condition and turn down requirements: the generated flash gas will be fully used by fuel system or recycled to the process gas upstream UNIT 310 without need to flare excess gas.

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1.2. DEFINITIONS AND ABBREVIATIONS

PCS (Plant Control System): It is a dedicated PLC used for control and operation of the facility.

SDV (Shut Down valve): On-off valve, usually with single effect pneumatic actuator.

BDV (Blow Down valve): On-off valve, usually with pneumatic actuator and with restriction orifice, used to depressurize a pressure boundary to the flare system in case of emergency.

LNG: Liquefied natural gas.

BOG: Boil off gas.

NRU: Nitrogen rejection unit.

ESD: Emergency Shutdown System.

PSD: Process Shutdown System.

TAL: Temperature alarm low.

TAH: Temperature alarm high.

FAL: Flow alarm low.

FAH: Flow alarm high.

PAL: Pressure alarm low.

PAH: Pressure alarm high.

LAL: Level alarm low.

LAH: Level alarm high.

TALL: Temperature alarm low low.

TAHH: Temperature alarm high high.

PALL: Pressure alarm low low.

PAHH: Pressure alarm high high.

LALL: Level alarm low low.

LAHH: Level alarm high high.

LO: Locked open.

LC: Locked closed.

P&ID: Piping and instrumentation diagram.

CSO: Car sealed open.

CSC: Car sealed closed.

PSV: Pressure relief valve.

TSV: Thermal safety valve.

LG: Level gauge.

LT: Level transmitter.

PI: Pressure indicator.

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2. REFERENCE DOCUMENTS

2.1. PROJECT DOCUMENTS

ITF Document N.	TAM Doc. N.	Document Title
PRO03-ING-FR01	--	Project Design Basis
PRO10-ING-PR01/02/03/04	21044-PR-PFD-00001	Process Flow Diagram
PRO10-ING-PR05/06/07/08/09/10/11	21044-PR-UFD-00001	Utility Flow Diagrams
PRO02-ING-FR01	21044-PR-HMB-00001	Heat and Material Balance
PRO13-ING-PR01	21044-PR-PID-20001	Piping and instrumentation diagram Inlet feed raw gas
PRO13-ING-PR02	21044-PR-PID-33001	Piping and instrumentation diagrams Sweetening unit
PRO13-ING-PR03	21044-PR-PID-33002	
PRO13-ING-PR04	21044-PR-PID-33003	
PRO13-ING-PR05	21044-PR-PID-33004	
PRO13-ING-PR06	21044-PR-PID-33005	
PRO13-ING-PR07	21044-PR-PID-33006	
PRO13-ING-PR08	21044-PR-PID-31001	Piping and instrumentation diagrams Dehydration unit
PRO13-ING-PR09	21044-PR-PID-31002	
PRO13-ING-PR10	21044-PR-PID-31003	
PRO13-ING-PR11	21044-PR-PID-35011	Piping and instrumentation diagram Cold Box
PRO13-ING-PR12	21044-PR-PID-35012	Piping and instrumentation diagram Nitrogen Rejection Unit
PRO13-ING-PR13	21044-PR-PID-35001	Piping and instrumentation diagram MR compressor package PK-351-1
PRO13-ING-PR14	21044-PR-PID-35002	Piping and instrumentation diagram MR compressor package PK-351-2
PRO13-ING-PR15	21044-PR-PID-35003	Piping and instrumentation diagram MR compressor package PK-351-3
PRO13-ING-PR16	21044-PR-PID-35004	Piping and instrumentation diagram MR compressor package PK-351-4
PRO13-ING-PR17	21044-PR-PID-35005	Piping and instrumentation diagram MR compressor package PK-351-5
PRO13-ING-PR18	21044-PR-PID-35006	Piping and instrumentation diagram MR compressor package PK-351-6
PRO13-ING-PR19	21044-PR-PID-35007	Piping and instrumentation diagram MR makeup
PRO13-ING-PR20	21044-PR-PID-35008	Piping and instrumentation diagram KO drum, compressors (simplified)
PRO13-ING-PR21	21044-PR-PID-35009	Piping and instrumentation diagram ORS

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PRO13-ING-PR22	21044-PR-PID-35010	Piping and instrumentation diagram Aircooled MR cooler, ammonia MR cooler, MR separator, MR pumps
PRO13-ING-PR23	21044-PR-PID-22001	Piping and instrumentation diagram LNG unloading skids
PRO13-ING-PR24	21044-PR-PID-22002	
PRO13-ING-PR25	21044-PR-PID-22003	Piping and instrumentation diagram LNG separator
PRO13-ING-PR26	21044-PR-PID-22004	Piping and instrumentation diagram LNG tank
PRO13-ING-PR27	21044-PR-PID-37001	Piping and instrumentation diagram Vaporizer skids
PRO13-ING-PR28	21044-PR-PID-41001	Piping and instrumentation diagram Hot oil unit
PRO13-ING-PR29	21044-PR-PID-36001	Piping and instrumentation diagram BOG compressor
PRO13-ING-PR30	21044-PR-PID-36002	Piping and instrumentation diagram BOG compressor
PRO13-ING-PR31	21044-PR-PID-36003	Piping and instrumentation diagram BOG compressor
PRO13-ING-PR32	21044-PR-PID-36004	Piping and instrumentation diagram BOG recycle compressor
PRO13-ING-PR33	21044-PR-PID-36005	Piping and instrumentation diagram BOG recycle compressor
PRO13-ING-PR34	21044-PR-PID-34001	Piping and instrumentation diagram Ammonia compressor package PK-341-1
PRO13-ING-PR35	21044-PR-PID-34002	Piping and instrumentation diagram Ammonia compressor package PK-341-2
PRO13-ING-PR36	21044-PR-PID-34003	Piping and instrumentation diagram Ammonia compressor package PK-341-3
PRO13-ING-PR37	21044-PR-PID-34004	Piping and instrumentation diagram Ammonia compressor package PK-341-4
PRO13-ING-PR38	21044-PR-PID-34005	Piping and instrumentation diagram Ammonia condensers, receiver, compressor simplified
PRO13-ING-PR39	21044-PR-PID-50001	Piping and instrumentation diagram Instrument air compressor / N2 generator
PRO13-ING-PR40	21044-PR-PID-41001	Piping and instrumentation diagram Hot oil unit buffer
PRO13-ING-PR41	21044-PR-PID-42001	Piping and instrumentation diagram Fuel gas conditioning
PRO13-ING-PR42	21044-PR-PID-43001	Piping and instrumentation diagram Diesel storage and distribution
PRO13-ING-PR43	21044-PR-PID-23001	Piping and instrumentation diagram Flare
PRO13-ING-PR44	21044-PR-PID-23002	
PRO13-ING-PR45	21044-PR-PID-23003	

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PRO13-ING-PR46	21044-PR-PID-55001	Piping and instrumentation diagram Closed drain
PRO13-ING-PR47	21044-PR-PID-53001	Piping and instrumentation diagram Oily water
PRO13-ING-PR48	21044-PR-PID-54001	Piping and instrumentation diagram Raw water
PRO02-ING-FR02	21044-PR-POD-00001	Process Description
SAF05-ING-FR01	21044-PR-RBP-00001	Relief & Blowdown Philosophy
SAF06-ING-FR01	21044-PR-HAC-00001	Hazardous area classification report
SAF12-ING-FR01	21044-PR-CEM-00001	C&E MATRIX

2.2. INTERNATIONAL CODES AND STANDARDS

NFPA Codes

Code No.	Title	Edition
10	Portable Extinguishers	2022
12	Carbon Dioxide Systems	2022
70	National Electrical Code	2020
72	Automatic Fire Detectors	2022
59A	Standard for the Production, Storage, and Handling of Liquefied Natural Gas (LNG)	2019

API Codes

Code No.	Title	Edition
RP 505	Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Zone 0, Zone 1, and Zone 2	2018
Std 520	Sizing, selection and installation of pressure relieving devices in refineries - Parts I and II	2020
Std 521	Pressure-relieving and depressuring systems	2020

IEC Codes

Code No.	Title	Edition
60079	Electrical apparatus for explosive gas atmospheres (all relevant parts)	2017

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IP Codes

Code No.	Title	Edition
3	Refining safety code	1981
15	Classification of Hazardous Areas	2015
19	Fire precautions at petroleum refineries and bulk storage installations	2012

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3. SAFETY

3.1. OBJECTIVES

The objectives of the Safety Philosophy are the following:

1. Ensure the safety in work environment for all the personnel in the facilities and in the neighbouring areas,
2. Identify the potential hazards generated by the Plant activities,
3. Plan prevention and protection measures to limit the effects of these potential identified hazards on the persons and on the work environment,
4. Ensure the safety of access and exits of the Plant for normal operations, maintenance or emergency.

The location of equipment throughout the plant has been optimised. Careful attention has been given to equipment location as this can improve the safety, accessibility and efficiency of the installation. Due consideration has been given to the location of process, utility, substations and control cabin as well as escape routes and safety equipment, having regard to the relevant hazards and risks.

3.2. MAIN FACTORS USED FOR PLANT LAYOUT CONFIGURATION

The general plant layout has been studied considering the following factors:

- Segregation of the zones on the basis of their risk content,
- Segregation of flammable hazardous sources from sources of ignition,
- Access and space for execution of maintenance, for fire-fighting operation and for emergency evacuation of site.

3.3. SAFETY DISTANCES

The principal spacing criteria for the installation relative to clearances between units is conceived according to the equipment spacing.

The flare will be installed at least at 100 m from the process and cabin areas.

3.4. CABINS

All cabins such as Control Room, Office cabins and other cabins will be installed in safe area inside air conditioned containers. A safe area (also referred to as unclassified or non-hazardous) is an area not

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classified as Zone 0, Zone 1 or Zone 2 according to API RP 505 criteria. This means that a flammable atmosphere is not expected to be present in the location.

3.5. EQUIPMENT AREAS

The areas where discharges of flammable liquid hydrocarbons could occur accidentally (vessels containing liquids having flanged connections) will be installed on concrete platforms having concrete containing curbs. Hydrocarbon spillages will be recovered manually by operator having portable pump and liquids recovery drum; dedicated procedures will be in place to avoid pollution during plant operation. Also all area containing LNG are provided with impoundment system in compliance with NFPA 59A.

3.6. PATHS FOR VEHICLES

Dedicated procedures together with safety signs are specifically issued for third parties transport means access to production area. The Italfluid operators and superintendent will supervise operations.

For security reasons third parties means access to site is restricted to few qualified and known contractor/persons. Refer to plant layout document CWO02-ING-PR01.

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4. CONTROL OF IGNITION: HAZARDOUS AREA CLASSIFICATION

The objective of controlling ignition hazard is to prevent any flammable releases from reaching possible ignition points.

4.1. HAZARDOUS AREA CLASSIFICATION

A hazardous area is a three-dimensional space in which a flammable atmosphere may be expected to be present in such frequencies as to require special precautions for the construction and use of electrical apparatus. All other areas are referred as to safe areas.

The hazardous area classification has provided the basis for design and classification of electrical equipment.

The hazardous area classification of the plant, the inlet manifold and transport systems of flammable substance has been performed in accordance with API RP 505.

The Hazardous areas have been classified as Zone 2. Two divisions will be considered as follows:

Zone 1	Location includes areas that are likely to have ignitable or flammable gases or vapours present under normal conditions.
Zone 2	Location is likely to have flammable gases or vapours present only under abnormal conditions.

The extension of hazardous areas will be calculated in compliance with the API RP 505. Refer to Hazardous area classification report and Hazardous area classification layout.

All electrical equipment for use in hazardous areas will be manufactured in compliant with the ATEX or NEC requirements for the hazardous area installation selected. ATEX execution is preferred above the NEC, but both options can be used.

4.2. VENTING SYSTEM

Vents that can release flammable fluids have been mainly directed to the flare as far as possible. Some LNG vents will be vented to atmosphere as permitted by NFPA 59A.

4.3. PUMPS

The shaft seals on pumps represent sources of leakage of fluids being pumped. Where these fluids are flammable or could cause environmental damage, the probability of leakage occurring will be minimised. This can be achieved by using high integrity shaft seals on the pumps. The most critical and hazardous pumps are the LNG pump on unloading skids. These pumps are seal less type installed inside a cryostat.

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5. SAFETY OF PROCESS AND UTILITY SYSTEMS

5.1. PROCESS SAFETY SYSTEMS

The installation has been designed and provided with devices that allow safe operation, control and shutdown for normal and abnormal conditions.

The process safety system is principally automatic, with the use of Emergency Shutdown system (ESD) and Fire & Gas system as well as mechanical relief systems, manual facilities for operator intervention and control will also be provided.

5.2. ISOLATION SYSTEMS

The following zones will be isolated:

- Gas treatment area: gas inlet (SDV-2001) until dehydration unit outlet (SDV-3560)
- Liquefaction area: from cold box inlet (SDV-3560) until LNG tank TA-221 (SDV-2201/2202) and BOG inlet (XV-3601)
- LNG tank loading bay

On critical control loops as per HAZOP recommendations, a block valve “fail safe SIL 2 (SDV)” will be installed, and the shutdown philosophy will be managed by the following levels:

1. PSD/LSD
2. ESD

5.3. RELIEF AND DEPRESSURIZATION SYSTEM

Gas treatment area will be equipped with block valves “fail close” (SDV-2001 and SDV-3560 and XV-3103A/B-3104A/B) and depressurized by means of “fail open” valves (BDV-2004, BDV-3101 and BDV-3561), both valve types managed by ESD system. Each blow downstream will be limited by restriction orifices.

The flare system will be designed for a full facility simultaneous blowdown, meaning it will be capable of handling the flow resulting from all valves failing open.

For process plant’s relief system the disposals are:

- Safety valve (PSV) for any pressure rating change, designed for full production conditions. One backup valve shall be installed if the equipment doesn’t have spare.
- Safety valve (PSV) designed to avoid vessel overpressure during fire case.

LNG storage tank relief system:

- LNG storage tank shall have local vent PSV collected with TSV stream on the tank roof.
- TSV-63116A shall be sent to IMPOUNDMENT AREA.
- LNG storage tank has 2x100% PSV designed for the worst design case venting the gas to atmosphere in a safe location. These valves are pilot operated and with proportional action.

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5.4. FLARES AND BURN PIT

Four (4) flares are forecasted for the same burning pit:

1. HOT FLARE collecting relief and blow down header. Effluents are coming from gas process area, from inlet (SDV-2001) to dehydration outlet (SDV-3560)
2. AMMONIA FLARE collecting ammonia (NH₃) only
3. COLD FLARE collecting relief and vents from COLD STREAMS. Effluents are coming from liquefaction area, from cold box inlet (SDV-3560) to LNG separator outlet (SDV-2201/2202)
4. SEME HOT FLARE collecting CLIENT'S blowdown, crossing the field through the same path of plant flares and ending at the same burning pit.

The burning pit will be realized with dike, at safety distance from plant facility. Proper fence will segregate the area to avoid human access.

The burning pit is realised on South direction because of prevailing wind direction blowing from North.

Hot flares will have flanged tip and ignition/continuous pilots with thermocouples for status remote supervision.

Continuous ammonia flaring is not foreseen in the operation of the plant. The Refrigeration unit (UNIT 340) works as a closed loop in such a way that ammonia will be flared only in case of emergency. Ammonia flare is properly designed to burn ammonia and is equipped with a pilot fuel line to ensure ignition. As stated above, the burning pit is located at safety distance from plant facility, being this an additional safeguard from ammonia toxicity affecting the operators.

5.5. LNG STORAGE TANK

SINGLE CONTAINMENT LNG storage tank, having internal wall insulated by perlite from external wall, with semi elliptic roof, having the following characteristics:

- Nominal capacity: 6020 m³
- Work capacity: 5652 m³
- Max. working pressure: 200 mbar
- The tank autonomy is sufficient for 10 days production at maximum LNG production rate.
- The LNG Tank is located on an impounding area according to NFPA 59A as follows:

Impounding areas serving one LNG container shall have a minimum volumetric holding capacity, V, that is one of the following:

- (1) *V = 110 percent of the maximum liquid capacity of the container*
- (2) *V = 100 percent where the impoundment is designed to withstand the dynamic surge in the event of catastrophic failure of the container*

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- (3) $V = 100$ percent where the height of the impoundment is equal to or greater than the container maximum liquid level

5.6. DRAINAGE AND SPILL CONTAINMENT

All process water drains shall be collected and sent to Process Water Treatment unit (UNIT 530).

Amine drains shall be sent to dedicated Exhaust Amine tanks having a recycle pump to the sweetening unit, or also with the possibility of being connected by hose to truck trailers for disposal.

OPEN AREAS: cockpits collecting rain water are forecasted for each platform realized with curbs. It's worth to highlight that being TENDRARA FIELD a gas treatment field no hydrocarbon spills are expected.

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6. EMERGENCY/PROCESS/LOCAL SHUTDOWN SYSTEM

6.1. GENERAL PRINCIPLE

The following intervention system are forecasted:

- Level I1, ESD: Emergency Shut Down in hazardous area
- Level I2, PSD: Process Shut Down
- Level I3, LSD: Local Shut Down
- Level I4, ASD: LNG loading bay Abandon Shut Down

Each shut down level is defined by its activation (inputs) and its actions (outputs).

ASD on LNG loading bay will be dedicated.

6.2. SYSTEM DESCRIPTION

Emergency shut down levels (ESD, PSD and ASD) are done within one integrated system which is fully independent from the process control system (PCS), as per API RP 14C recommendation.

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Level	Function	Description
I1	ESD	PROCESS AREA AND LNG LIQUEFACTION SHUT DOWN
		<i>Manual activation by:</i> ESD push buttons ESD buttons along escape ways Ref. to escape ways and ESD pushbuttons layout
		<i>Automatic activation:</i> Process areas and loading bay fire detection LNG spill detection in Tank area
		<i>Effects:</i> Closing of Inlet and outlets of the plant Plant depressurization Main power generation shut down with consequent start of emergency diesel generators and emergency users and panel activation (see emergency panel power balance) Output signal to Owner's facilities (to be defined) Refer to SAF12-ING-FR01 C&E MATRIX.
I2	PSD	PROCESS AREA AND LNG LIQUEFACTION SHUT DOWN
		<i>Manual activation by:</i> PSD push button
		<i>Automatic activation:</i> Process area combustible and ammonia gas detection Process parameter (LAHH, PAHH, LAHH, TAHH, etc) out of set point FLARE no ignited pilot on flare tip Power plant shut down Instrument air pressure low on IA header Hot oil unit shut down Ammonia Refrigeration System shutdown MR Refrigeration System shutdown Smoke detection in cabins, Fire detection in non-hazardous Area
		<i>Effects:</i> Closing of Inlet and outlets of the process and LNG liquefaction plant. Some units may still be running. Refer to SAF12-ING-FR01 C&E MATRIX.

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I3	LSD	<i>Package shut down logic implemented on PCS for:</i> Ammonia refrigeration Unit Mix refrigerant Unit BOG Unit 360 LNG pump skids
		<i>Manual activation by:</i> Local push buttons
		<i>Automatic activation</i> <i>Separator function LSLL that stops pumps or closes liquid valve outlet</i> Single packages logic (Ammonia refrigeration unit, Mix refrigerant unit, BOG compression unit, LNG pump skids, LNG loading bay, Hot oil unit, el.power generation unit. LAHH 3111 on E311 sweet gas precooler LALL 3103 on V311 low level shut down on liq, separator upstream dehydration, LT3105 HH and LL on V312 regeneration wet gas separator TAHH 3102 on TT regeneration gas feed TALL 480x on each F.G. line of Power Generator PAHH 2201 on LNG tank PT 3702 - PAHH on E-371 LNG vaporizer
		<i>Effects:</i> Refer to SAF12-ING-FR01 C&E MATRIX.
I4	ASD	LNG LOADING BAY ABANDON SHUT DOWN
		<i>Manual activation by:</i> Local push buttons
		<i>Automatic activation:</i> Loading bay area gas detection <i>LNG spill detection (temperature probe)</i>
		<i>Effects:</i> Closing of XV2211- XV2212- XV2217 SDV pump 221A Closing of XV2213- XV2216- XV2214 SDV pump 221B Vapor line is protected from backflow by check valve Closing of XV2221- XV2222- XV2227 SDV pump 222A Closing of XV2223- XV2226- XV2224 SDV pump 222B Vapor line is protected from backflow by check valve

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The following detection system shall be adopted:

- **IR Flame Detectors** are installed to detect the fire around compressors and pumps, process vessels/skid mounted units and valves in process area. Main view area is summarised in the following Table. For each detector, location and height shall be decided during construction stage based on Table.

Protected equipment	Main View Area
Compressor	Main flanges and coupling area
Pump	Flanges, coupling area, and surrounding ground
Vessel	Lower part flanges and surrounding ground
Valve, etc	Lower part flanges and surrounding ground

IR Flame Detector shall detect the presence of an approximate 0.09 m2 fire, within and angle of view 90° to 120° at a distance of 15 m from the detector.

IR flame Detector shall be located to maximize the field of view.

IR flame Detector shall not generate fault alarm caused by sunlight or other intense light source. The view angle of IR Flame Detectors shall be adjusted to avoid the effect of flare flame in the construction stage.

FIRE ZONES	
Process area	
200	Feed gas inlet receiving unit
220	LNG storage, metering system and loading station unit
230	Flare and burn pit unit
310	Dehydration unit
330	Sweetening unit
340	Refrigeration unit
350	Liquefaction unit
360	Flash gas compression unit (PK-361 and PK-362)
370	Vaporizer and fuel gas unit
410	Hot oil unit
500	Instrument/Plant air generator unit
530	Process oily water treatment unit
540	Raw feed water Treatment unit

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600	Nitrogen generator unit
Utilities area	
480	Diesel Electric Generator unit
	Diesel storage and supply unit
	Gas Engine Electric Power Unit /
	Electrical substation
730	Firefighting units
900	Cabins
1000	Living camp

Alarm Horns

Alarm horns shall be installed so that the sound can be easily heard normally at the height of approximately 2,1 m above ground. **Different tone shall be used for each different cause of alarm. Some selected horns around LNG tank shall be located at higher elevation as below, so that alarm sound can be audible inside the tank dike. Refers to document SAF07-ING-PR02.**

Combustible Gas Detector

Each combustible gas detector installation height will be decided in the construction stage based on the following Table.

Protected Equipment		Service	Detection Point
Main plant	Pump	Liquid	Near the pump at the height of approximately 0.5 m above ground or floor
	Compressor	Light Gas	Two detectors shall be located near, above and on both sides of the main flanges
	Control Valve	Light gas	1.0-3.0 m above and near control valve
		Heavy Gas	Approximately 0.5 m above ground or floor and near control valve
		Liquid	
	Gas Engines	Light Gas	Near and above fuel gas inlet flange of engine enclosure

Each point gas detector shall be basically located within 3 m from the detection point. When one detector is installed to the protected equipment, this detector shall be located in the prevailing wind direction.

References apply also to document SAF11-ING-FR01 – F&G positioning criteria.

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6.3. FIRE AND GAS DETECTION LAYOUT AND LOGIC

- Gas process area zone 1:** inlet gas (SDV 2001) until dehydration unit (SDV 3560)

UNIT	GAS DET.
200	CH ₄
310	
330	

- Gas process area zone 2:** from Cold Box (SDV 3560) to LNG Separator 221 (SDV 2201/2202)

UNIT	GAS DET.
221	
350	
370	

- Loading bay**

UNIT	Safety device	Case of	
S-221/S-222	CH ₄ det	LNG spillover with no ignition of fire	ASD I4 (no impact on the process) – LNG spill will be collected in the LNG impoundment where foam pourer and deluge valve will be activated by Low Temperature sensors
	Fire det.	LNG spillover and ignition of fire	ESD and manual CO ₂ extinguisher intervention locally. LNG spill will be collected in the LNG impoundment where foam pourer and deluge valve will be activated by Low Temperature sensors. Emergency power generation black start

- Gas engines for power generation**

UNIT	Case of	Case of	Effect/action
480	CH ₄ det		Alarm Signal to Control Cabin
		Fire det.	Plant ESD and manual CO ₂ extinguisher Emergency power generation black start

- Cabins**

UNIT	Safety device	Case of	Effect/action
MCC	Smoke detectors (2 of 2)	fire	Alarm Signal to Control Cabin Plant ESD and use of manual CO ₂ extinguishers
Control room	Smoke detectors (2 of 2)	fire	Alarm Signal to Control Cabin Plant ESD and use of manual CO ₂ extinguishers
Office cabins	Smoke detectors (2 of 2)	fire	Alarm Signal to Control Cabin use of manual CO ₂ extinguishers

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7. ACTIVE FIRE PROTECTION

7.1. GENERAL PRINCIPLES

Fire protection system is designed on the following basis:

- Control and extinguishment of only one of fire per time, occurring across the whole plant area;
- Fire spreading is assumed not to exceed the fire area limits in which it rises;
- Design water demand shall not exceed the greatest amount of water required for each fire scenario.

7.2. FIRE PROTECTION PRINCIPLES

Fire protection of equipment will be achieved by means:

- Application of low expansion foam to control flammable and combustible liquid hydrocarbons having a flash point below 65°C;
- Application of dry chemical powder to control small scale gas fires;
- Application of spray water to control fire radiation to adjacent equipment;
- Total flooding by CO2 extinguisher agent of normally unmanned enclosure;
- Total flooding by clean agent of normally manned enclosure.

7.3. SELECTION OF EXTINGUISHING/COOLING AGENTS

- Spring well filtered water shall be used as fire water;
- Foam concentrate shall be AFFF (AQUEOUS FILM-FORMING FOAM) suitable for dilution with water;
- Dry chemical powder shall be class B-C type for application to hydrocarbons in plant outdoor areas.

7.4. EXTINGUISHING/ COOLING AGENTS APPLICATION RATES

Figures marked with @ are in compliance with NFPA codes.

Application of low expansion foam for control of liquid hydrocarbon fires (rates are expressed as litres of water)

- 10.0 l/min/m² for fixed roof tank
- 10.2 l/min/m² for vessels
- 20.4 l/min/m² for pumps

Application of Spray Water for protection of equipment from “exposures” and equipment “Exposed to radiation”

- 10.2 l/min/m² for vessels
- 36.0 l/min/m² for atmospheric fixed roof storage tanks (not applicable)
- 20% of total tank shell flow rate for roof of fixed roof tank (not applicable)
- 20.4 l/min/m² for pumps
- 40.0 l/min/m² for compressors

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7.5. FIRE WATER NETWORK SPECIFICATIONS

The water storage capacity is the amount of water stored and dedicated to fire service only. The minimum storage of water shall be based on 8 hours supply period.

The fire water circuit shall be pressurized with two electrically driven jockey pumps (one spare).

The fire water network shall be designed assuming water flowing through all branches at same time and considering water demand at each reference areas.

Pipe sizing shall be performed assuming 3.0 m/s maximum water velocity (on main network) at the water demand of the area under examination.

Generally the network shall be located underground, deep enough to avoid winter freezing and located along the less dangerous side of plant.

The main ring will be verified by hydraulic calculation in order to have a minimum pressure of 7.0 barg at and at the farthest take-off point of the area.

The fire water piping will be installed as a looped system, with strategically located isolation valves.

7.6. FIRE PUMPS SPECIFICATIONS

Two main fire water pumps, centrifugal type, will be installed.

Both pumps shall be diesel engine driven; the capacity of each one shall be 100% of maximum fire water demand and will ensure maximum fire-water supply even if one of the pumps is out of service or when the electric power is not available.

The main water pump shall be controlled by a low low pressure signal from the fire water distribution network. After a calibrated delay time should, for any reason, the fire water ring pressure be lower than its set point, jockey pump returns the pressure to set point value.

Pumps can also be manually started from pump local panel.

7.7. FIRE WATER SYSTEM FOR LOGISTIC AREA

The fire water system shall be designed to run at the required flow rate and designed pressure for all the requested period.

The minimum flow rate for the farthest hydrant shall be 500 l/minute and pressure equal to 7.0 barg.

Two (2) hydrants at the same time shall be considered in use, in system design.

7.8. FOAM SYSTEM

Concentrate foam will be stored in a centralised storage vessel, to allow the charging of foam from lorry tanks and foam barrels.

Foam composition is made by mixing AFFF (Aqueous Film Forming Foam) at 3% in water.

Foam protection system

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The system provides the supply of premixed foam, solution formed at local stations (pressure proportioning with diaphragm) each one provided with the following characteristics:

- Foam concentrate storage (bladder tank)
- Foam concentrate refilling nozzle
- Foam concentrate proportioning
- Foam solution delivery valves

Number and capacity of foam station shall be determined on the basis of:

- Foam system location
- Foam system requirements

Concentrate foam station will be provided for:

- Hydrocarbon liquid storage
- Plant process unit

Concentrate foam for LNG storage

The foam concentrate bladder tank storage capacity shall allow supply foam of 55 min. for fixed roof tank.

Concentrate foam for plant process unit

The foam concentrate bladder tank storage capacity shall allow supply foam for 30 min. for each fire area where the fire could be considered confined.

7.9. FIRE PROTECTION SYSTEMS AREA/ VOLUME METHOD APPLICATION

The following fixed systems shall be provided for dedicated fire protection of equipment/area as indicated below:

7.10. LOW EXPANSION FOAM SYSTEM (TOP TANK SURFACE AND FOAM DELUGE SYSTEM)

They shall be provided for protection of:

- Atmospheric fixed-roof tank (manually)
- Process area

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7.11. WATER SPRAY SYSTEM

Water spray system (through fixed distributors) shall be manually and/ or automatically operated as indicated below:

- Vessel (manually/ automatically)
- Columns (manually/ automatically)
- Gas compressors & pumps (manually/ automatically)

Water spray systems shall be designed to guarantee the rate for protection of equipment exposed to radiation.

Carbon dioxide total flooding system

CO₂ based total flooding systems shall be provided for protection of:

- Normally unmanned enclosure

Systems shall be automatically operated upon fire alarm signal from proper fire detection section and shall be provided with interlock, which impedes the discharge when a man is inside the enclosure.

7.12. FIRE PROTECTION EQUIPMENT FOR LOCAL APPLICATION

Fixed equipment

The following fixed equipment are provided through the plant for local application of extinguishing agent.

Fire Hydrants

The hydrants will be connected with the fire main and located at no more than 50m each other in process area, in the utility and storage area.

The hydrants shall be 6" (150 mm) in diameter. Each hydrant will be provided with two connections 2½" in diameter for hose service, one 4" connection for fire truck service and one upper flange 4" for application of closed flange monitor.

Hydrants points should be positioned so that any fire can be reached by two hoses from different hydrants.

Piping for fire water supply should be protected against any damage.

Fire Hose Box

Hose boxes will be installed near each hydrant.

Each hose box shall consist of reinforced fibre glass cabinet containing two hoses having 2½" in diameter and 25 m in length, plus two straight/ fog flow nozzle.

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Fire Water Monitors

Fire water monitors, fixed type, will be installed generally on hydrants and will be equipped with a manually adjustable nozzle capable to discharge up to 2000l/ min at 7.0 barg

Fire water monitors shall be located in the process area.

A minimum distance of 15 m from the protected equipment shall be provided and are protected from possible road traffic accident.

Each fire monitor shall have rotation angle of 360° and elevation angle of 80°. The action range shall be of a minimum of 40m in horizontal straight.

Hose Reel

Hose reel connected to fire water network will be installed along the pipe rack and near compressor area at 50m each one.

Each hose reel will be equipped with 30 m of 1" not collapsible hose, water nozzle, 1" globe valve and cabinet.

The jet shutting with a nominal delivery of 1000 l/min with a supply pressure of 7 barg.

Please refer to document SAF02-ING-FR04 for dedicated firefighting equipment specification following the firewater and foam design.

Deluge Valve System

Deluge valves will be utilised for automatic or manual intervention of water and foam water deluge system in LNG storage tank area and loading bay area.

In process areas in case of fire detection the deluge system is not present. Operators will activate the relevant foam monitor to unit on fire and providing only cooling by water application on the surrounding units by means of different monitors and hydrants in the closest area.

7.13. MOBILE EQUIPMENTS

Firefighting mobile equipment are part of the firefighting layout document SAF02-ING-PR01 and will include the following:

- Foam trailers with premixer (trailers to be connected to FF hydrants on one side; foam to fire chambers to the other).
- Monitors trailers.

More detailed information is given on spec SAF02-ING-FR04.

Extinguishers

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The extinguisher agent depends on fire type:

- Pressurized portable foam extinguishers suitable for fires type A-B.
- Wheeled extinguishers 50 kg dry chemical and 30 Kg CO₂ suitable for fires A-B-C.
- Portable extinguishers 12 kg dry chemical and 5 Kg CO₂ suitable for fires B-C.
- Portable extinguishers 6 kg dry chemical and 3 Kg CO₂ suitable for fires B.

Extinguishers will be located in strategic places in well discernible locations:

- In process and utilities units, portable powder extinguishers will be hung on pipe rack and on access stairways such that the travel distance from any part of process area to one extinguisher does not exceed 15 meters. In addition, and according to the unit area one or two wheeled powder extinguishers are supplied.
- In addition to the on-plant extinguishers a number of extinguishers are stored for immediate replacing those exhaust. Number, location, size and type of fire extinguishers, including the spare number in store warehouse shall be on Contractor's fire fighting drawings.

Please refer to document SAF07-ING-PR02, Safety Equipment Lay-out.

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8. PASSIVE FIRE PROTECTION

Passive Fire Protection has been applied as a complement to the Active Fire Protection. The main features of passive protection are below underlined:

1. Equipment spacing is a first passive protection: the minimum distance between process equipment belonging to different process areas is established to segregate the fire to spread into other fire areas (refer to “Plant Layout with equipment spacings” and “minimum equipment spacing table”).
2. Platform kerbs are a second passive protection: they will segregate the fire to spread into other process areas.
3. Plant lay-out conception is a third passive protection: lay-out has been designed to keep flammable storage are away from process area, and utilities (Power generators, control room, etc) area away from flammable areas (process, containment, etc).

Passive fire protection insulation is used in oil refineries or industrial process plants to minimize the escalation of fire that could result from structural failure and overheating of pressure vessels.

Due to the presence of the ESD system that depressurizes the process equipment in case of fire, the fire source will be eliminated. Structural collapse will not consequently occur because of limited time exposure to fire, so passive fire protection insulation is not required in the facilities.

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9. ESCAPE AND EVACUATION

9.1. SITE ACCESS-EMERGENCY VEHICLES

Adequate access for emergency service vehicles to enter and circulate around the site will be provided.

The largest envisaged emergency service vehicles and their access requirements have been considered in the plant layout. In fact, particular attention has been paid to the turning radio required by these vehicles in the installation.

9.2. PERSONNEL ACCESS

Doors

Rapid means to escape of the cabins will be foreseen. All cabins will be equipped with antipanic doors to emergency exits.

Doors will generally be orientated so that they open in the direction of escape.

Doors on escape routes shall not be capable of being locked and will be provided with self-illuminated signs.

Ladders

Maximum continuous ladder height will be 9 metres. Where the ladder height exceeds this, rest landings will be placed at a maximum of 6 metre intervals.

9.3. EVACUATION AND ESCAPE WAYS

Rapid means of escaping from all of the cabins and the plant structures will be provided. The plant will also be provided with evacuation routes and emergency exits. At least two evacuation routes shall be provided for every location in the Plant.

These evacuation routes will be never obstructed and their orientations have been arranged in different directions so that, in case of danger, the personnel can always move away and make their way to a safe area.

Access to structures has taken into account of needs for regular visits and maintenance, the requirements for equipment handling and of mobile cranes.

Walkways and platforms in process areas have been designed for safe escape. Equipment layout, permits walkways and platforms to be laid out with a minimum of dead ends and give sufficient space at high levels for safe execution of all routine maintenance operations.

In the event of power failure due to a Diesel Power Generator default, the Control Cabin, the MCC cabin and the other 2 cabins will be illuminated continuously by emergency lights because they are under UPS Supply (Uninterruptible Power Supply). Moreover, the UPS will supply power to:

- n°1 illumination pole located at the centre of the installation
- n°3 emergency light located at the 3 entrance doors

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- n°2 emergency lights located on the Power Generators front panels

9.4. SAFETY SIGNS AND NOTICES

Safety signs will be provided throughout the plant to provide warning or information, or to identify locations of safety, fire-fighting and survival equipment, etc.

They will combine the geometrical shape, colour and pictorial symbol to give specific health/safety or emergency information, instruction, prohibition and/or advice on safety matters.

In accordance with BS 5378 safety signs are divided into four main categories as follows:

Meaning	Shape	Colour
Prohibitive signs	Circular	Red band with red cross band
Obligation sign (Mandatory sign)	Circular	Blue background White contrast
Warning sign	Triangular	Yellow background Black contrast
Emergency Information	Rectangular	Green background White contrast

Safety signs and notices having text will be written in French and Arab languages.

Emergency signs will be provided to inform personnel of escape routes, emergency exits, fire-fighting and rescue equipment locations.

Emergency signs will be located in accordance with good engineering practice.

Alarm System

The Process Control System is implemented to display an alarm on the screen and to provide a distinct sound in case one operating parameter overcomes a threshold value (minimum/maximum), established for each set-point.

The Audible Alarm sound audible in Control room is typically differentiated between pre-alarm and alarm as follows:

pre-alarms: TAL, TAH, FAH, FAL, PAH, PAL, LAL, LAH. They are generated through a separate buzzer (high frequency sound), installed in the control room and connected to the Process Control System. The sound is intermittent. The alarm is reset manually on the PCS display (software reset button).

alarms: TALL, TAHH, PAHH, PALL, LAHH, LALL. They are generated through a separate buzzer (low frequency sound), installed in the control room and connected to the Process Control System. The sound is intermittent. The alarm is reset manually on the PCS display (software reset button).

The facility will also be provided with a General audible alarming system that will rapidly alert the personnel present inside the installation. This alarming system will be activated during emergency situations by means of a local siren above the roof of the control room.

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The General Alarm sound audible in all the points of the installation is typically differentiated between PSD, ESD, Forced Evacuation and Gas Leakage. The general Alarm system is managed by the PCS and will be powered from UPS batteries.

PSD Alarm Sound (inlet SDV valve is shut-in due to an abnormal upset of condition that led to generate an alarm and sub-sequent interruption of incoming fluid). A pitched sound is produced intermittently by the siren and remains audible until the process parameter that caused the PSD sequence is restored inside the operating range and the alarm is manually reset on the PCS display (software reset button).

Forced Evacuation: A low tone sound is produced intermittently by the siren and remains audible until the alarm is reset manually on the PCS display (software reset button).

Eventually, another source of audible alarm is the ESD pneumatic panel. A pneumatic siren is located on the panel 100-CP-001 and alerts the operators in case of an ESD sequence activated. A pitched sound of a different tone is produced continuously until it is interrupted manually by operator, once the operating conditions are restored.

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10. PERSONNEL PROTECTION

10.1. PERSONAL SAFETY EQUIPMENT

Suitable safety helmets have to be available for use on the plant. Sufficient protective clothing (eye protectors, ear protectors, goggles, gloves, overalls and safety boots) will be available for all personnel engaged in operations where they are exposed to risk of injury.

Portable flammable gas detectors will be available, together with oxygen depletion monitors where needed.

All equipment provided will be kept clean and in a good state of repair and condition and be readily available for use when required by a responsible person.

10.2. AUXILIARY FACILITIES

Emergency Eye Wash and Shower

Safety showers will be provided in all areas where accidental contact with chemicals is possible, especially in chemical storage areas.

They will be equipped with eye wash fountains. They will be readily accessible and immediately available in the event of an emergency.

They will not be obstructed or obscured from view in all directions. Where visual obstructions cannot be avoided, means shall be provided to indicate their locations. They will use potable water only. After use the standpipe will drain automatically.

10.3. MACHINERY GUARDING

All exposed moving parts on all machinery will be suitably guarded to protect personnel from injury. Guards and safety devices provided will be well maintained and kept in position whilst machinery is running. They should only be removed under a work permit.

10.4. PERSONNEL PROTECTION AGAINST HOT AND COLD SURFACES

Reasonable safeguards are to be adopted to eliminate the possibility of accidents caused by personnel coming into contact with either hot or cold surfaces which will impart sudden partial shock or injury.

Where surfaces are deliberately exposed to emit heat or perform cooling functions, suitable identification will be provided to give personnel adequate warning.

The plant operators and any personnel working inside the plant shall use personal safety devices before entering the plant process area and operating any type of equipment.

The plant operation manager will develop a plant safety policy and specify details of the personal safety devices required by the personnel entering or working inside the plant.

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All surface operating at a temperature not higher than 250°C and not requiring insulation for good operation are not insulated and not protected against accidental contact because all personnel working inside the plant shall use personal safety devices as above specified.

It is preferred to avoid using insulation if not strictly required to have better accessibility and maintenance space around the equipment.

Equipment and piping operating at a temperature above 250°C, not subject to cold or hot insulation for good operation may require personal protection insulation. When personal protection is required, physical barriers such as open mesh guards, protective metal sheeting/screens or hand railings and hazard markings shall be used instead of insulation.

In order to minimize the possibility of operators coming into contact with hot or cold surfaces, each skid is designed to permit routine monitoring from the boundary, without need of entering the skid.

10.5. MEDICAL FACILITIES

A medical and first aid nurse will be provided in the plant area along with an infirmary cabin equipped with necessary medical devices, medicines and drugs.

10.6. EMERGENCY SAFETY SHOWERS AND EYE BATHS

In areas where personnel may come into contact with hazardous chemicals, eye baths will be provided. Where larger quantities of hazardous chemicals are being handled a combined safety shower and eye bath have been provided.

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11. EMERGENCY POWER

The power generation is provided by n°2 diesel Power generator in configuration 2 x 100%. In case of operational unit failure, the start-up of the stand-by unit can be manual or automatic. The automatic option is configurable on the local PLC resident inside the parallel system control panel. The parallel system control panel is installed to operate the periodical switch from the operational unit to the stand-by unit without interruption of power supply.

An emergency power system is provided by an UPS unit. In case of breakdown of the emergency electricity supply system, a system of batteries supplies the following principal systems for a period at least 30 minutes.

- Plant Control System,
- Gas and smoke detection system,
- ESD command panel,
- Some emergency lights outside cabins and Power generation units.

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12. ISOLATION

12.1. OBJECTIVES

The objectives of the Isolation Philosophy are the following:

- To identify and describe isolation methods for online plant containing hazardous or not hazardous fluids so that maintenance/inspection can take place with minimum plant shutdown.
- To reduce the requirement for periods of plant shutdown for maintenance/inspection.
- To outline facilities required for system/equipment draining, flushing and venting.

12.2. METHODS OF ISOLATION

Various methods of isolation are described in this section and illustrated in Appendixes II to IV.

Single Block Valves

Single block valves will provide the balance of general isolation as follows:

- For isolation, including that of instruments, in process lines and utility services (whether hazardous or not).
- For vessels, columns and exchangers where maintenance intervals are relatively long (minimum 12 month).
- In lines at recognised limits of operation for example rundown line to storage tank.
- For fill, vent, drain and bleed connections on system pipework not used during normal operation and fitted with a blank flange.
- On filter and small pressure vessel where is not required to enter inside for maintenance or routine operation.

Double block valves

This method is used only for small lines (1½" and smaller), and only for positive isolation of pressure vessels.

Removable spools

Removable spools are used to provide an air gap for isolation. This isolation method involves removing a spool section from the piping and installing a blind flange to block the flow from the source.

Spades/Spectacle blinds

For all drains sizes 2" and bigger, spectacle blinds shall be provided along with an isolation valve, as shown in APPENDIX III.

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Spectacle blinds may also be used for equipment isolation as an alternative to single block valves or removal of spools.

Locked Valves

For safety reasons, certain block valves are designated 'locked open' (LO) or 'locked closed' (LC) on the P&IDs. For operational reasons, some valves may be 'car sealed open' (CSO) or 'car seal closed' (CSC).

In general, the use of locks and car seals for valves is to be minimised. Isolation valves shall therefore only be locked or car sealed in position (either open or closed) if inadvertent operation of the valve would lead to damage to equipment or an unsafe condition. This is to keep the use of Car Seals and Locks to a reasonable minimum to avoid these devices becoming de-valued through frequent use.

Specific applications requiring locked or car sealed valves are listed below together with the type of locking device to be used.

Locking devices will be used for:

- PSV isolation valves.
- Manual valves in vent lines to flare.
- Main isolation valves in the firewater headers and foam systems.
- Valves which must be open to guarantee a relief route (e.g. block valves which must be opened to allow for thermal expansion).
- Block valves in pump suction lines not rated for discharge pressure.
- Block valves downstream of BDV's.
- Manual blowdown lines, e.g. production flowlines blowdown.
- SDV bypasses.
- Suction/discharge valves of multiple pumps if one pump is on autostart.

Car seals will be used for:

- Block valves downstream of spill-off control valves used for routing excess gas to flare.
- Valves in drain lines from diesel and methanol day tanks.
- Isolation valves for pressure and level trip instrumentation.
- Isolation valves for differential pressure instrumentation having electronic interlocks.
- Isolation valves in pump recycle/minimum flow lines.
- Block valves in Potable Water lines to safety showers and eyebaths.

It should be noted that valves in the following services will not be locked or have car seals:

- Level bridle isolations.

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SDV Valves

SDV valves may be used as isolation block valves. This will be reviewed on a case by case basis.

Pressurisation bypasses around SDV valves, where specified, shall be locked closed or be itself an SDV.

12.3. VESSEL ENTRY – ISOLATION REQUIREMENTS

Isolation of tanks, vessels, etc. before entry requires positive isolation of the confined space from all potential hazards. The method involves the removal of spools from all inlet and outlet connections to the confined space, or, as an alternative, the use of spades/spectacle blinds. Physical breaks (e.g. removable spools, use of spades/spectacle blinds) shall occur as close to the vessel as possible. Blind flanges of the correct specification shall be applied to all open pipe ends where possible. Relief valves are a type of removable spool piece. They shall be removed and a blind flange installed on the open pipe ends to isolate the vessel from the flare header.

12.4. RELIEF VALVE ISOLATION

Introduction

Safety relief valves protecting items of equipment and systems are required to be inspected at regular intervals to ensure safe operation and may require replacement at any time if they start to leak. The affected equipment or system would have to be taken out of service to replace or maintain a single relief valve. This is undesirable particularly for major items of equipment and systems required for continuous operation. Therefore spare relief valves shall be installed in locations where it is unacceptable for an item of equipment or system to be shutdown before the relief valve can be maintained. Block valves for relief and blowdown pipelines (i.e. in the headers) shall be provided with a purpose-built locking device, for locking valves open. Parallel slide or gate valves fitted on the duty should be installed in the inverted position so that the valves tend to fail to the open position.

It is then necessary to ensure:

- i) that it is impossible to isolate the operating and spare relief valves at the same time.
- ii) that there is an unobstructed path through to flare on the operating valve.
- iii) that the relief valves available will be adequately sized to discharge at the required capacity for the equipment or system.

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Spared Relief Valves

Where a relief valve is spared, all outlet isolation valves will be locked open, while the inlet isolation valve to on-line relief valve will be secured in the fully open position, and the inlet isolation valve to the spare relief valve will be locked closed.

For the on-line relief valve, LO inlet and outlet valves ensure a clear relief flow path.

The spare relief valve inlet valve shall be LC in order to avoid an inadvertent opening leading to both relief valves relieving simultaneously with possibility of chattering and internal damage.

A proper procedure shall be placed to describe the operational sequence to be carried to change from the on-line relief valve to the spare relief valve. This procedure must ensure that one of the valves is available for equipment protection at all times. In case of using a changeover valve instead of the inlet isolating valves, the changeover valve itself always assures that one of the two ways is open. Downstream the PSVs, both outlet valves shall be LO.

Maintenance of relief valves will generally be undertaken during shutdown periods after depressurisation, therefore single valves upstream and downstream are adequate.

Provision of Single Relief Valve Only

For the following cases, provision of single relief valves only may be considered. No outlet valve is required if the discharge is to atmosphere or open drain.

- i) For systems or equipment which can be safely taken offline and isolated without disturbing production. No inlet block valve. Lock open outlet.
- ii) For systems or equipment when the only relief case to be considered is fire. No inlet block valve. Locked open outlet.
- iii) For relief valves which protect against thermal expansion. No inlet block valve. Locked open outlet.

12.5. CONTROL VALVES

Normally control valves will be provided with a by-pass.

For fail closed valves, drain connections shall be installed on each side of the control valve inside any block valves. For fail open or fail in position valves, a drain valve shall be provided on the upstream side of the control valve only.

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12.6. PROCESS AND INSTRUMENT CONNECTIONS

Vents and Drains

All vessels shall be fitted with a valved vent connection for maintenance purposes.

A drain valve will be provided either on the vessel or on the liquid outlet line to permit full draining of the vessel before breaking containment.

Utility Connections

Where connections are required for equipment/system inerting they shall be clearly marked on the P&I diagrams. The purge size and location(s) shall be selected to ensure that the system can be fully purged within a reasonably short period of time.

Check valves shall be provided upstream of block valves in the case of hard piped utility connections to prevent backflow and contamination of the utility. Where not hard piped, a blinded block valve shall be provided (Non-return valves will also be required in all utility services connected directly with process equipment).

Instrument Isolation

Isolation for instruments shall be essentially as per other guidelines contained in this document.

General requirements for flow and pressure instruments are given in Appendix IV.

Isolation requirements for level instruments is covered in Appendix IV.

Development of the isolation philosophy for instruments, particularly level instruments, will require detailed attention. Consideration should be given to each type of instrument (i.e. displacer, dp cell etc., for level) with respect to criticality of service, calibration and maintenance requirements.

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APPENDIX I – SYMBOLS KEY



Block valve



Spectacle blind point

LT

Level transmitter

PDX

Differential pressure transmitter or differential pressure gauge

PX

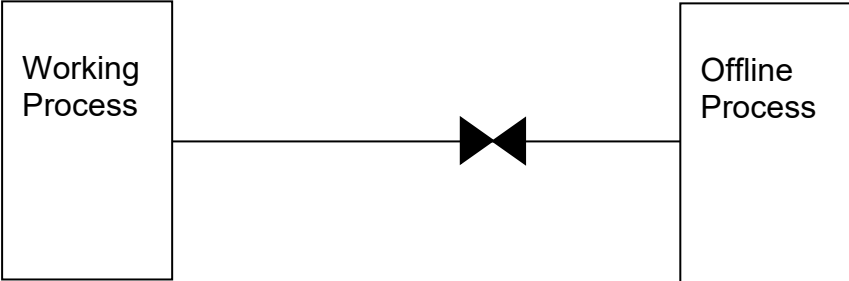
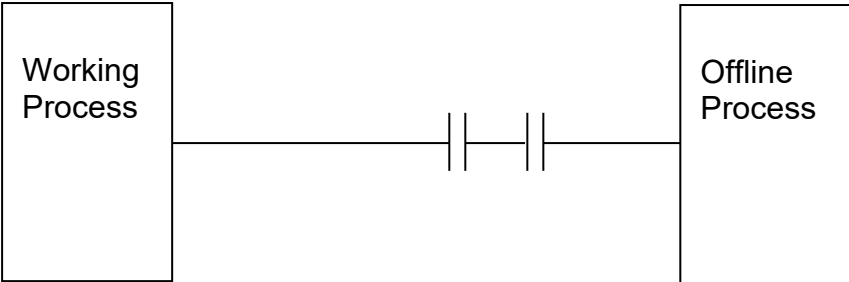
Pressure transmitter or pressure gauge



Flanges

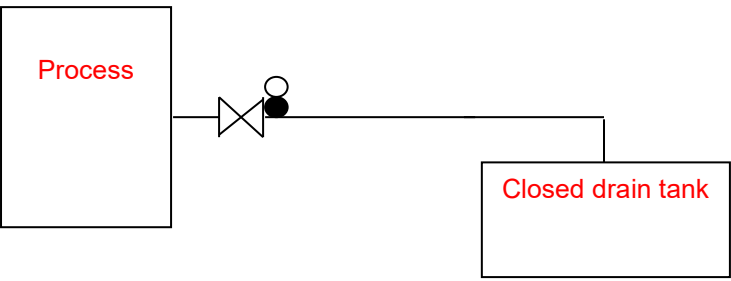
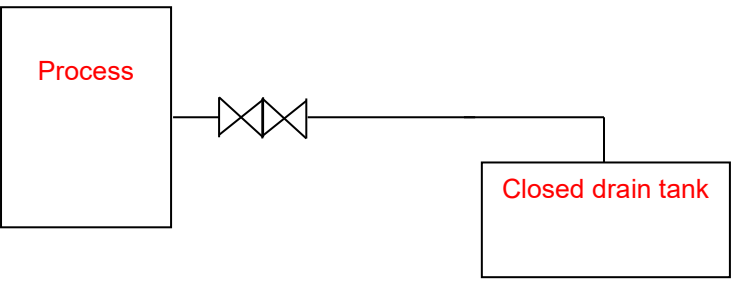
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APPENDIX II – EQUIPMENT ISOLATION

Application	Diagram
<u>SINGLE BLOCK VALVE</u>	
<u>REMOVABLE SPOOL</u>	

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APPENDIX III – EQUIPMENT DRAINS

Application	Diagram
2" and bigger drains	
1½" and smaller drains	

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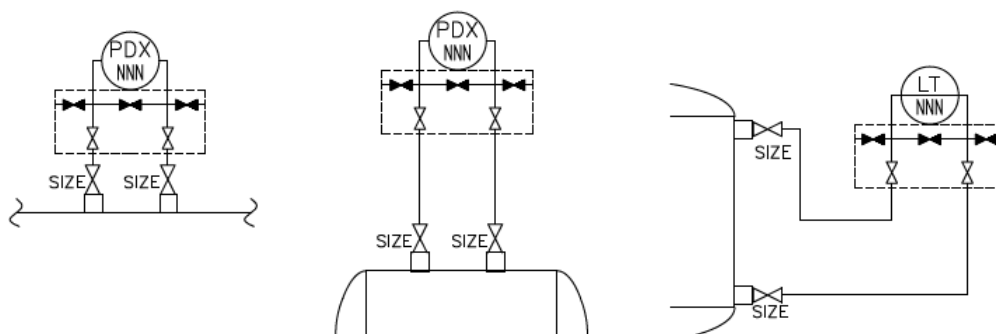
APPENDIX IV - INSTRUMENTATION (PRESSURE, FLOW, LEVEL, ETC.)



Root valves are used when the instrument is local mounted, for example pressure transmitters or pressure gauges.



For remote mounted pressure transmitters or pressure gauges, root valves are used along with block and bleed manifolds.



For differential pressure transmitters or gauges and level transmitters, root valves are used along with 5-way block and bleed manifolds.

On low pressure lines in cryogenic services (LNG) other considerations are applied to avoid overpressure due to thermal expansion so root valves may not be used on remote mounted instruments.